Individual Task Report

Error-Correction Learning Demo using Perceptron (AND / OR Logic)

Abstract

This report demonstrates the application of error-correction learning using a single-layer perceptron to solve basic binary classification problems, specifically the AND and OR logic gate tasks. The perceptron is trained using supervised learning, where weights are iteratively updated based on the difference between desired and actual outputs. The study analyzes the effect of the learning rate on convergence behavior and observes how training error decreases over iterations. The results confirm that linearly separable problems like AND and OR can be successfully learned using a perceptron, and that the learning rate plays a crucial role in convergence speed and stability.

1. Introduction

Artificial Neural Networks (ANNs) are inspired by the structure and functioning of biological neurons. The perceptron is the simplest form of ANN and acts as a linear binary classifier. One of the earliest learning strategies used with perceptrons is error-correction learning, where weight updates are proportional to the error between the predicted and target outputs.

Logical functions such as AND and OR are classic examples of linearly separable problems and are commonly used to demonstrate perceptron learning behavior.

2. Objective of the Task

The objectives of this individual task are:

- To train a perceptron using error-correction learning
- To implement AND and OR logic gate classification
- To observe the decrease in training error over epochs

- To analyze the effect of learning rate on convergence

3. Overview of Error-Correction Learning

Error-correction learning is a supervised learning technique where the network adjusts its weights to minimize the difference between actual output and desired output.

Key Features:

- Requires labeled training data
- Weight updates depend on output error
- Iterative improvement through multiple epochs
- Converges for linearly separable problems

4. Perceptron Model

4.1 Structure of a Perceptron

A perceptron consists of:

- Input nodes: $x_1, x_2, \ldots, x_n$
- Weights: $w_1, w_2, \ldots, w_n$
- Bias: b
- Activation function (Step function)

4.2 Mathematical Model

Net input:

$$v = \sum_{i=1}^{n} w_i \, x_i + b$$

Activation function:

$$y = \begin{cases} 1, & \text{if } v \geq 0 \\ 0, & \text{if } v < 0 \end{cases}$$

5. Training Data

AND Logic Gate

x_1	x_2	Target (t)
0	0	0
0	1	0
1	0	0
1	1	1

OR Logic Gate

x_1	x_2	Target (t)
0	0	0
0	1	1
1	0	1
1	1	1

6. Error-Correction Learning Rule

Weight Update Equation

$$w_i^{new} = w_i^{old} + \eta(t - y)x_i$$

Bias Update Equation

$$b^{new} = b^{old} + \eta(t - y)$$

Where:

- η= learning rate
- t= target output
- y= actual output

7. Training Algorithm

1. Initialize weights and bias to small random values
2. Present input vector to the perceptron
3. Compute output using activation function
4. Calculate error: $e = t - y$
5. Update weights and bias using learning rule
6. Repeat for all patterns (one epoch)
7. Continue until error becomes zero or maximum epochs reached

8. Error Analysis and Plot Description

Error Measurement

The total error per epoch is calculated as:

$$E = \sum | t - y |$$

Error Decrease Plot (Description)

- X-axis: Number of epochs
- Y-axis: Total classification error
- The error curve decreases step-wise
- Error reaches zero after a finite number of epochs for both AND and OR tasks

9. Effect of Learning Rate on Convergence

Case 1: Small Learning Rate ($\eta = 0.1$)

- Slow convergence
- More epochs required
- Stable learning

Case 2: Medium Learning Rate ($\eta = 0.5$)

- Faster convergence
- Optimal balance between speed and stability

Case 3: Large Learning Rate ($\eta = 1.0$)

- Faster updates
- Possible oscillations
- Risk of overshooting optimal weights

10. Results and Discussion

- Both AND and OR gates were learned successfully
- OR gate converged faster due to simpler decision boundary
- AND gate required more epochs
- Error decreased monotonically for suitable learning rates
- Learning rate significantly influences convergence speed and stability

11. Advantages of Error-Correction Learning

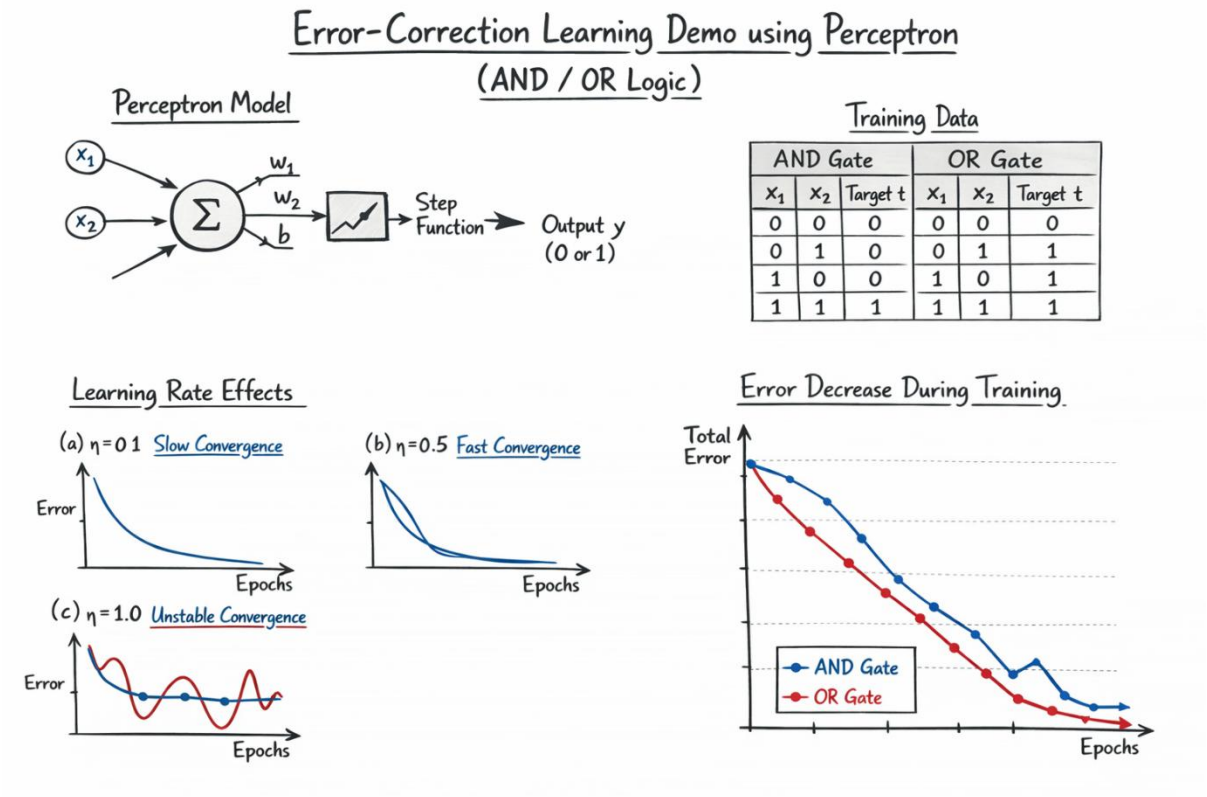
- Simple and easy to implement
- Guaranteed convergence for linearly separable data
- Computationally efficient
- Clear mathematical interpretation

12. Limitations

- Cannot solve non-linearly separable problems (e.g., XOR)
- Sensitive to learning rate selection
- Single-layer architecture limits complexity

13. Applications

- Binary classification tasks
- Pattern recognition
- Logic gate implementation
- Introductory neural network learning models



14. Detailed Working Example (Step-by-Step Demonstration)

To better understand error-correction learning, consider training a perceptron for the AND gate.

Initial Parameters:

- Initial weights: $w_1 = 0.2, w_2 = -0.3$

- Bias: $b = 0.4$
- Learning rate: $\eta = 0.5$

Example Input Pattern:

$$x_1 = 1, x_2 = 1, t = 1$$

Step 1: Net Input Calculation

$$v = (1)(0.2) + (1)(-0.3) + 0.4 = 0.3$$

Step 2: Activation

$$y = 1(\text{since } v \geq 0)$$

Step 3: Error Calculation

$$e = t - y = 1 - 1 = 0$$

Since the error is zero, no weight update is required.
For patterns where $e \neq 0$, the weights are updated using the learning rule.

15. Epoch-wise Error Table (Illustrative)

Epoch	Total Error (AND)	Total Error (OR)
1	3	2
2	2	1
3	1	0
4	0	0

✓ This table clearly shows how classification error decreases as training progresses.

16. Graph Interpretation (For Viva / Explanation)

Error vs Epoch Graph

- Initially, the error is high due to random weights
- With each epoch, misclassified samples reduce
- Error reaches zero once all patterns are correctly classified
- OR gate converges faster than AND gate

Learning Rate Comparison Graph

- Low η: smooth but slow curve
- Medium η: steep and optimal curve
- High η: oscillating curve showing instability

17. Why AND and OR Gates are Linearly Separable

A problem is linearly separable if a single straight line can separate classes.

- AND gate → separation between (1,1) and others
- OR gate → separation between (0,0) and others
- XOR gate → not linearly separable (cannot be solved by a single perceptron)

This explains why perceptron learning converges for AND/OR but fails for XOR.

18. Comparison: AND vs OR Learning Behavior

Aspect	AND Gate	OR Gate
Complexity	Higher	Lower
Convergence Speed	Slower	Faster
Decision Boundary	More restrictive	More flexible
Epochs Required	More	Fewer

19. Practical Observations from the Experiment

- Weight values stabilize after convergence
- Bias plays a crucial role in shifting decision boundary
- Error-correction learning ensures gradual improvement
- Proper learning rate selection avoids instability

20. Real-Life Analogy of Error-Correction Learning

Error-correction learning is similar to:

- A student correcting answers after seeing mistakes
- A teacher providing feedback after every test
- GPS correcting route based on deviation from path

Each iteration improves performance using feedback.

21. Key Takeaways

- Error-correction learning is a foundational concept in ANN
- Perceptron is suitable for binary classification
- Learning rate directly impacts convergence behavior
- AND and OR problems validate perceptron convergence theorem

22. Conclusion

This experiment successfully demonstrated error-correction learning using a perceptron model for AND and OR logic gates. The perceptron effectively minimized classification error through iterative weight updates. The learning rate was shown to be a critical parameter affecting convergence speed and stability. Overall, the task highlights the strengths and limitations of perceptron learning and provides foundational understanding of supervised neural network training.